

LTE Features and Installation in India

Long-term evolution (LTE) technology will change the way we connect and communicate. Read on to find out when and how



(Image courtesy: www.fiercetelecom.com)

RINKI SHARMA

Though LTE is popularly called the 4G wireless communication technology, it is not the name of a technology. Long-term evolution or LTE is a Third Generation Partnership Project (3GPP).

3GPP is a collaboration between groups of telecommunication associations. It has given us well-known technologies like global system for mobile communications (GSM), general packet radio service (GPRS), enhanced data rates for GSM evolution (EDGE), universal mobile telecommunication system (UMTS), high-speed packet access (HSPA), and recently LTE and LTE Advanced (LTE-A).

The project on LTE started in November 2004, with an aim to enhance 3GPP's already existing UMTS by simplifying the existing hybrid circuit-switched/packet-switched core network with a packet-only core network. This helps in achieving higher data rates leading to higher throughput,

lower latency leading to lower delay or faster data transfer, lower communication cost and enhanced system efficiency. The requirements for LTE or system architecture evolution are reduced cost per bit, reduced power consumption by terminals, flexible use of new or existing frequency bands, and simplified network architecture.

LTE is bound to be effective for both the consumers and the service providers/operators. It promises to provide data rates of up to 100 Mbps for downlink and 50 Mbps for uplink, within a cell of 5km radius. The number of users that can be supported within the cell can vary from 200 to 400. Channel bandwidths can vary from 1.4 MHz to 20 MHz in both uplink and downlink. LTE also claims to support mobility speeds of up to 350 kmph with inter-cell soft handovers. It promises high performance for mobile speeds of up to 120 kmph and functional support for speeds of 120 kmph to 350 kmph. Support for 350 kmph to 500 kmph is under consideration. Since LTE has evolved from already-existing 3G technologies, operators can upgrade the existing infrastructure to support LTE, which will prove to be economical and efficient for the operators.

How will LTE change the way we communicate?

It is easy to observe the difference that LTE is going to make in our daily lives. With support for high data rates, high mobility speeds

and soft-handover, we would be able to have voice and video chats on the move apart from attending to official work on the Internet. LTE will also prove useful in infotainment and telematics applications.

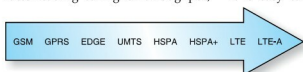
When is LTE arriving in India?

According to the latest news updates, India will use 2.3 GHz of the spectrum for time-division LTE (TD-LTE). The broadband wireless access spectrum auction for TD-LTE has been won by operators including Reliance, Airtel, Aircel and Tikona and chip manufacturer Qualcomm. Swedish telecom equipment maker Ericsson has won the LTE network infrastructure and management deal in India with Indian telecom company Augere Holdings awarding the contract to Ericsson. Telecom operator Airtel and telecom equipment maker Huawei have announced completion of their field trials of LTE-TDD in 2011. Very recently, mobile handset manufacturer Nokia has announced that it is ready to support LTE when it arrives in India.

Going by the predictions of industry experts, by mid-2012, LTE data-enabled devices may be launched in the market and by 2013-15 we may get LTE-based Internet telephony services on smartphones. Bharti Airtel has already launched 4G LTE services in Bengaluru and Kolkata. This high-speed wireless broadband network is based on TD-LTE.

With all the benefits that LTE would provide, India is waiting to brace LTE for better connectivity. ●

The author is working as an assistant professor at M.S. Ramaiah School of Advanced Studies, Bengaluru, and pursuing Ph.D in wireless networks from Coventry University, UK



Evolution of 3GPP standards